ARTIFICIAL INTELLIGENCE – ENCS3340

Faculty of Engineering & Technology

Electrical & Computer Engineering Department

Comparative Study of Image Classification Using Decision Tree, Naive Bayes, and Feedforward Neural Networks

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Date: 28 / 6 / 2025

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1. Introduction

In the era of Artificial Intelligence, image classification stands as a crucial task within computer vision. This report compares the classification performance of three models: Naive Bayes, Decision Tree, and Multilayer Perceptron (MLP). All models are trained on a custom grayscale image dataset of birds, horses (cavallo), and jellyfish. We evaluate the models based on their accuracy, interpretability, and learning capacity, supported by confusion matrices and decision visualizations.

2. Dataset Description and Preprocessing

- Total images: 704
- Classes:
  - Bird (Label 1)
  - Cavallo (Horse) (Label 2)
  - Jellyfish (Label 3)

Each image was:
- Converted to grayscale
- Resized to 64×64
- Flattened into a 4,096-dimensional vector
- Normalized (for MLP)
- Reduced via PCA (50 components) for Naive Bayes and Decision Tree

Data was split into 80% training and 20% testing using stratified sampling.

Table 1 : Data sets

Dataset	Description
D1	Balanced small dataset (dog=88, bird=131, cat=166) — all cat images are faces only.
D2	Random images, large in size (bird=139, cat=126, dog=134).
D3	Similar to D2 with different sampling (bird=138, cat=129, dog=126).
Dall	Combination of D1, D2, and D3 (bird=321, cat=328, dog=253).
DX1	Large dataset (horse=2623, cat=1668, dog=4863) — bird class renamed to horse.
DX2	Same as DX1 but cat class renamed to sheep (sheep=1820).

3. Model Implementation

3.1 Naive Bayes Classifier

Used GaussianNB from scikit-learn on PCA-reduced data. Chosen for its simplicity and baseline performance. Does not require scaling.

3.2 Decision Tree Classifier

Used DecisionTreeClassifier with:
- criterion='entropy', max_depth=35, min_samples_split=4, min_samples_leaf=2
Applied on PCA-reduced data. Offers hierarchical decisions and clear interpretability.

3.3 Feedforward Neural Network (MLPClassifier)

Used MLPClassifier with:
- hidden_layer_sizes=(1024, 512, 256)
- max_iter=2000, early_stopping=True, validation_fraction=0.1
Trained on scaled original data. Best at learning complex, nonlinear relationships.

4. Results Per Model

Naive Bayes

Accuracy: 0.70
Baseline performance; struggles with feature dependencies.

Decision Tree

Accuracy: 0.70
Improved interpretability, comparable accuracy to Naive Bayes.

Decision Tree (First 3 Layers):

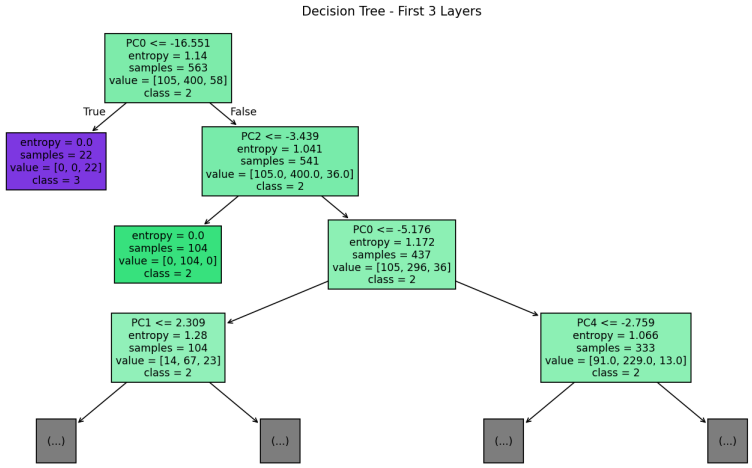

Figure 1: Decision

Neural Network (MLP)

Accuracy: 0.83
Confusion Matrix:
[[8 15 3]
 [1 96 3]
 [0 2 13]]

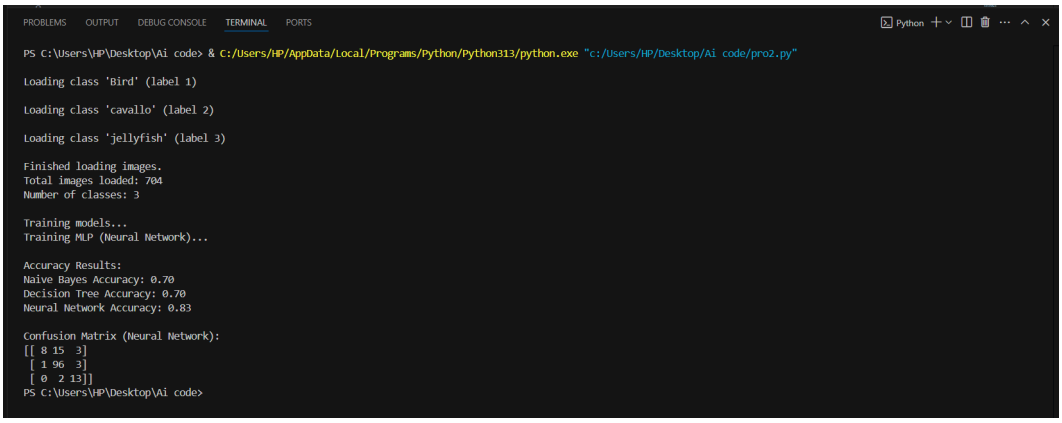

Figure 2 : Output

5. Comparative Analysis

Model	Accuracy
Naive Bayes	0.70
Decision Tree	0.70
Neural Network	0.83

- Naive Bayes: Limited by independence assumption.
- Decision Tree: Understandable model structure, but prone to overfitting.
- MLP: Strongest performer due to nonlinear learning, though more computationally intensive.

6. Discussion and Conclusion

6.1 Strengths and Limitations

- Naive Bayes: Fast and simple, but ignores feature correlations.
- Decision Tree: Easy to visualize, but sensitive to depth.
- MLP: Best accuracy, handles complexity, needs more resources.

6.2 Challenges

- Tuning MLP parameters
- Visual similarity between classes in grayscale
- Balancing depth in decision trees

6.3 Future Work

- Try CNNs for spatial feature learning
- Use data augmentation to improve robustness
- Explore hyperparameter tuning (e.g., GridSearchCV)
- Use color images or additional features (e.g., HOG)

Conclusion:

This study highlights the balance between interpretability and performance in image classification. While Naive Bayes and Decision Trees offer simplicity and visualization, the Neural Network significantly outperforms both in accuracy, showcasing the power of deep learning on image data.

Test cases :

```
/Library/Frameworks/Python.framework/Versions/3.12/lib/python3.12/site-pack
ransparency expressed in bytes should be converted to RGBA images
  warnings.warn(
Naive Bayes Accuracy: 0.42
Decision Tree Accuracy: 0.53
Neural Net Accuracy: 0.45
Confusion Matrix:
[[ 9  7  7]
 [ 7 17  6]
 [ 7  3 15]]
yousefshanti@yousefs-MacBook-Air pro2_file % /usr/local/bin/python3 /Users/
/Library/Frameworks/Python.framework/Versions/3.12/lib/python3.12/site-pack
ransparency expressed in bytes should be converted to RGBA images
  warnings.warn(
Naive Bayes Accuracy: 0.59
Decision Tree Accuracy: 0.69
Neural Net Accuracy: 0.62
Confusion Matrix:
[[26  7  0]
 [ 6 17  4]
 [ 6  1 11]]
```

```
  warnings.warn(
Naive Bayes Accuracy: 0.62
Decision Tree Accuracy: 0.72
Neural Net Accuracy: 0.27
Confusion Matrix:
[[30  1  1]
 [ 8 10  3]
 [ 8  1 16]]
yousefshanti@yousefs-MacBook-Air pro2_file % /usr/local/
/Library/Frameworks/Python.framework/Versions/3.12/lib/p
 bytes should be converted to RGBA images
  warnings.warn(
Naive Bayes Accuracy: 0.69
Decision Tree Accuracy: 0.73
Neural Net Accuracy: 0.44
Confusion Matrix:
[[28  1  5]
 [ 3 11  5]
 [ 3  4 18]]
```

```
Naive Bayes Accuracy: 0.61
Decision Tree Accuracy: 0.65
Neural Net Accuracy: 0.49
Confusion Matrix:
[[18  9  5]
 [ 3 24  4]
 [ 5  1  8]]
yousefshanti@yousefs-MacBook-Air pro2_file % /usr/lo
/Library/Frameworks/Python.framework/Versions/3.12/
 ConvergenceWarning: Stochastic Optimizer: Maximum i
  warnings.warn(
Naive Bayes Accuracy: 0.62
Decision Tree Accuracy: 0.61
Neural Net Accuracy: 0.51
Confusion Matrix:
[[20  6  3]
 [ 6 20  4]
 [ 5  6  7]]
```

```
Naive Bayes Accuracy: 0.47
Decision Tree Accuracy: 0.59
Neural Net Accuracy: 0.38
Confusion Matrix:
[[17  5  2]
 [ 7 16  3]
 [10  5 14]]
yousefshanti@yousefs-MacBook-Air pro2_file %
```